2D KS Tests

PHYS 4730 Lab 17
Fall 2026

Setup



- In 1987, a supernova was observed in the Large Magellanic Cloud.
- Among other things, 19 elementary particles known as *neutrinos* were detected from that direction. For each, a **time t_i** and **energy E_i** was recorded.
- On the basis of this data, a model was developed to explain neutrino production in supernovae.
- Now, today a second supernova has been observed. We would like to compare this to the 1987 event, and the theoretical model to which it led.

Setup

- I remember learning in PHYS 4730 about the KS Test for comparing distributions of a single RV. Is there something similar for multiple RVs?
- I find a paper in an old *Computers in Physics* journal that looks like it matches the problem I'd like to solve.
- Strategy – reproduce the paper's result, then we can be confident in applying the code to our supernova neutrino problem.

Kolmogorov-Smirnov Test For Two-Dimensional Data

William H. Press and Saul A. Teukolsky

How to tell whether a set of (x,y) data points are consistent with a particular probability distribution, or with another data set

The Kolmogorov-Smirnov (K-S) test is a standard statistical test for deciding whether a set of data is consistent with a given probability distribution. For example, the data set might be the list of values $x = 0.021$, 0.082, 0.143, 0.520, 1.788, and 4.712, and the probability distribution might be $p(x)dx = e^{-x}dx$. The classical K-S test is one-dimensional–each data point is a single number–and the above example is a one-sample test, where one data set is compared to an analytic model. There is also a similar two-sample K-S test, which has two data sets instead of a data set and a model. The two-sample test checks whether two data sets are both drawn from the same underlying probability distribution, but without assuming any specific model for that distribution. For example, the two-sample K-S test could be used to determine whether the list of values above could come from the same distribution as the list $x = 0.054$, 0.129, 0.384, 2.452. Note that our two data sets need not have the same size. In practice the K-S test would be done with a larger number of data points in each set, no less than of order 10 or 20.

These one-dimensional K-S tests are described in full detail in most statistics textbooks. While it would be redundant to give a complete discussion here, a quick summary of the mechanics of these one-dimensional K-S test will be a necessary preface to our main subject, which is a two-dimensional generalization of the K-S test.

Any model distribution function $p(x)$ has an associated cumulative distribution function

$$P(x) \equiv \int_{-\infty}^{x} p(s)ds.$$

Obviously, since probablity is normalized, this varies from 0 at $x = -\infty$ to 1 at $x = \infty$. A data set x_i, $i = 1...N$ also has a cumulative distribution function, namely the "staircase function" $S_N(x)$ defined as the fraction of the N data values to the left of x. Note that $S_N(x)$ starts at zero, jumps by the same constant $1/N$ at each x_i, and finishes at 1.

The Kolmogorov-Smirnov statistic is the maximum difference between the cumulative distribution function of the data and model (one-sample), or between the two data sets (two-sample), that is,

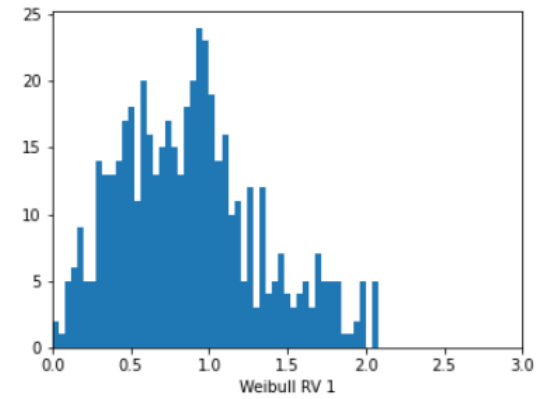
$$D = \max_{-\infty < x < \infty} |S_N(x) - P(x)| \quad (1)$$

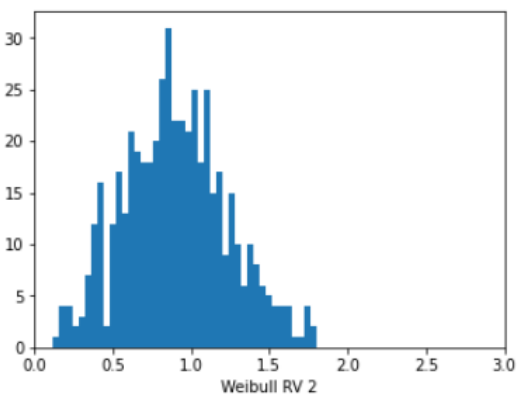
$$D = \max_{-\infty < x < \infty} |S_{N_1}(x) - S_{N_2}(x)| \quad (2)$$

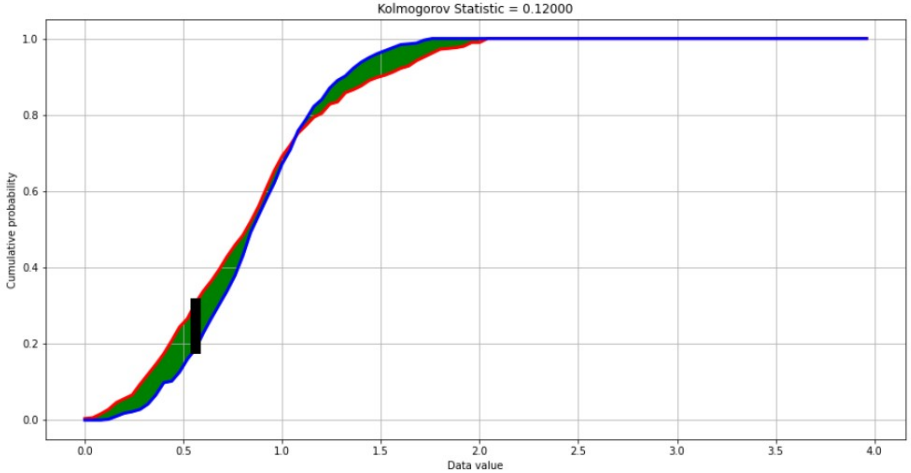
What makes the K-S statistic useful is that its distribution in the case of the null hypothesis (data sets

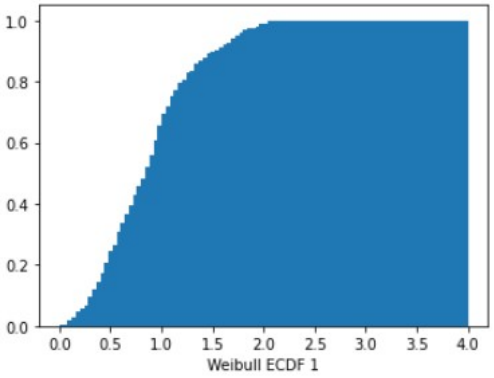
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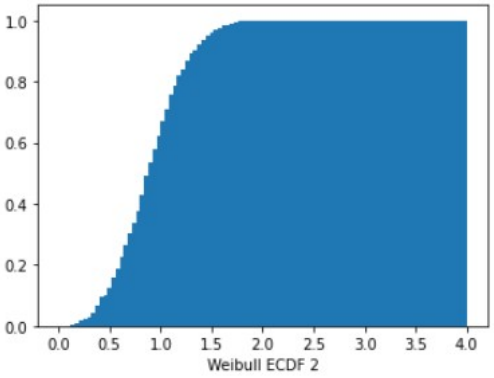
Recap: *p*-value from KS Statistic











$$N = \frac{nm}{n+m}$$

$$p = P(D_N \geq d_{obs}) = 2\sum_{k=1}^{\infty}(-1)^{(k-1)}e^{-2k^2Nd_{obs}^2}$$

Need to do some code conversion…

```
      SUBROUTINE QUADCT(X,Y,XX,YY,NN,FA,FB,FC,FD)
        Given an origin (X,Y), and an array of NN points with coordinates XX and YY, count how
        many of them are in each quadrant around the origin, and return the normalized frac-
        tions. Quadrants are labeled alphabetically, counterclockwise from the upper-right. Used
        by KS2D1S and KS2D2S.

      DIMENSION XX(NN),YY(NN)
      NA=0
      NB=0
      NC=0
      ND=0
      DO 12 K=1,NN
        IF(YY(K).GT.Y)THEN
          IF(XX(K).GT.X)THEN
            NA=NA+1
          ELSE
            NB=NB+1
          ENDIF
        ELSE
          IF(XX(K).GT.X)THEN
            ND=ND+1
          ELSE
            NC=NC+1
          ENDIF
        ENDIF
       12 CONTINUE
      FF=1.0/NN
      FA=FF*NA
      FB=FF*NB
      FC=FF*NC
      FD=FF*ND
      RETURN
      END
```

FORTRAN

```python
### function to return normalized fraction in each quadrant ###

### Given an origin (X, Y), and an array of NN points with  ###
### coordinates XX and YY, count how many of them are in    ###
### each quadrant around the origin, and return the         ###
### normalized fractions. Quadrants are labeled             ###
### alphabetically, counterclockwise from the upper-right.  ###

def quadct(x,y,xx,yy):

    nn = len(xx)

    na,nb,nc,nd = 0.0,0.0,0.0,0.0

    for k in np.arange(0,nn):
        if(yy[k] > y):
            if(xx[k] > x):
                na += 1.0
            else:
                nb += 1.0
        else:
            if(xx[k] > x):
                nd += 1.0
            else:
                nc += 1.0

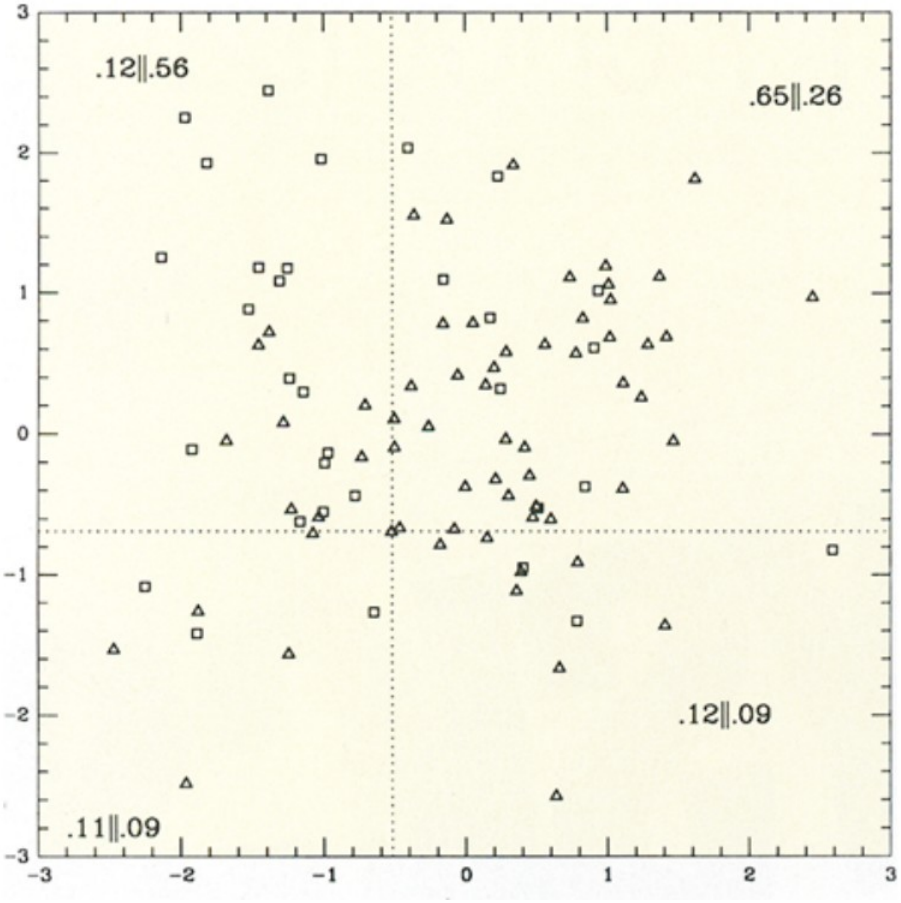
    fa = na/nn
    fb = nb/nn
    fc = nc/nn
    fd = nd/nn

    return fa,fb,fc,fd
```

python

Today's Lab

- 2D Kolmogorov-Smirnoff Test
- Goal: Duplicate Press & Teukolsky result
- JB provides points. *Caveat*: Apparently in P&T analysis they included some points that were beyond the edges of their plot!



Final Comments

- HW04 P2 should be easy after today's lab.
- I didn't specify the number of points for the Gaussians. In principle your code should work for any number, but N = 100 is what the instructor used.